Kimi Delta Attention

Deep Dive into Kimi K3 — Chapter 4

Yin Yang

Foundation Books Project

The one idea of this chapter

KDA replaces a growing token cache with a fixed-size memory
that is *corrected*, not just written to —
and must be chunked without changing the ordered recurrence.

The one idea of this chapter

KDA replaces a growing token cache with a fixed-size memory
that is *corrected*, not just written to —
and must be chunked without changing the ordered recurrence.

Goal: derive the update from a reconstruction loss, add channel-wise retention, carry the complete layer state across calls, and prove chunkwise evaluation exact — then keep numerics, cost, and evidence in their own ledgers.

Roadmap

From additive memory to the delta rule

The recurrence and the complete layer

The ten-token trace and chunking

Numerics, cost, and evidence

From additive memory to the delta rule

Predict, then correct

- Additive memory $S \leftarrow S + kv^\top$ superposes: writing $u_1 \mapsto (0, 1)$ over $u_1 \mapsto (1, 0)$ reads back $(1, 1)$.

Predict, then correct

- Additive memory $S \leftarrow S + kv^\top$ superposes: writing $u_1 \mapsto (0,1)$ over $u_1 \mapsto (1,0)$ reads back $(1,1)$.
- One gradient step on $\frac{1}{2}\|R^\top k - v\|^2$ gives $R + \beta\, k(v - R^\top k)^\top$: correct only what is wrong.

Predict, then correct

- Additive memory $S \leftarrow S + kv^\top$ superposes: writing $u_1 \mapsto (0, 1)$ over $u_1 \mapsto (1, 0)$ reads back $(1, 1)$.
- One gradient step on $\frac{1}{2}\|R^\top k - v\|^2$ gives $R + \beta\, k(v - R^\top k)^\top$: correct only what is wrong.

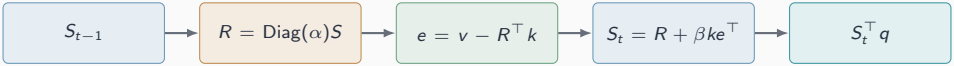
β is a write strength, not a token probability; the step changes activation memory, not learned weights.

The recurrence and the complete layer

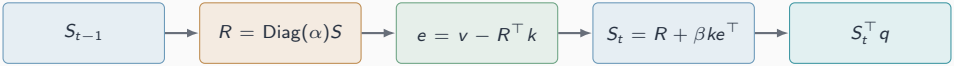
Retain, predict, correct, read



Retain, predict, correct, read



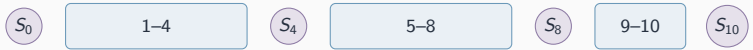
Retain, predict, correct, read



Channel-wise α gives each key row its own lifetime; the complete layer state is the matrices *plus* three convolution histories (44 elements in the toy layer).

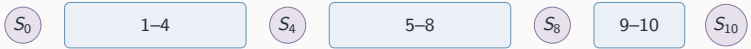
The ten-token trace and chunking

Same boundaries, either way



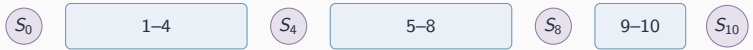
- Affine segments compose in order: $(\Phi_2\Phi_1,\ \Phi_2\Psi_1 + \Psi_2)$.

Same boundaries, either way



- Affine segments compose in order: $(\Phi_2\Phi_1,\ \Phi_2\Psi_1 + \Psi_2)$.
- A strict triangle builds corrections; an inclusive triangle gives reads. Chunk sizes 1–12 all reproduce the trace exactly.

Same boundaries, either way



- Affine segments compose in order: $(\Phi_2\Phi_1,\ \Phi_2\Psi_1 + \Psi_2)$.
- A strict triangle builds corrections; an inclusive triangle gives reads. Chunk sizes 1–12 all reproduce the trace exactly.

Restoring the matrices but not the histories gives a different continuation with the same shapes.

Numerics, cost, and evidence

Bounded decay, separate ledgers, narrow claims

- K3 bounds log-decay to $(-5, 0)$: a 16-position tile reciprocal stays below e^{80}, inside BF16 range.

Bounded decay, separate ledgers, narrow claims

- K3 bounds log-decay to $(-5, 0)$: a 16-position tile reciprocal stays below e^{80}, inside BF16 range.
- State is fixed; work is linear with serial span $\Theta(N)$; formal FLOP crossover $T_\star = 496$ — not a latency claim.

Bounded decay, separate ledgers, narrow claims

- K3 bounds log-decay to $(-5, 0)$: a 16-position tile reciprocal stays below e^{80}, inside BF16 range.
- State is fixed; work is linear with serial span $\Theta(N)$; formal FLOP crossover $T_\star = 496$ — not a latency claim.
- Synthetic tasks, ablations, and operator speed each support one narrow inference.

Bounded decay, separate ledgers, narrow claims

- K3 bounds log-decay to $(-5, 0)$: a 16-position tile reciprocal stays below e^{80}, inside BF16 range.
- State is fixed; work is linear with serial span $\Theta(N)$; formal FLOP crossover $T_\star = 496$ — not a latency claim.
- Synthetic tasks, ablations, and operator speed each support one narrow inference.

Finite state cannot address every token — Chapter 5 adds the global-attention path.